Supermarket Case Study: Student Steps

Oct 7, 2026

Work through these steps in order on `supermarket_sales_case_study.csv` (two years, three branches). Order matters: drop exact duplicates before resolving conflicts, and fix negative quantities before checking `Total`.

Cleaning

1. **Load and look.** Read the file and check how many rows and columns it has, the first few rows, and each column's type.
2. **Audit before you fix.** Count the missing values in each column, the rows that are exact copies, and the rows that share an Invoice ID. List the distinct values in Gender and Payment.
3. **Drop exact duplicates.** Rows identical in every column are a resent upload. Remove the copies.
4. **Resolve conflicting Invoice IDs.** Some IDs appear twice with different Quantity and Total. The later row is a correction, so keep the last one. Write this rule down as a comment.
5. **Standardise text columns.** Remove extra spaces and lowercase, then map each value to one correct spelling: Male / Female and Cash / Credit card / Ewallet.
6. **Fix negative quantities.** A minus sign is a typing error. Formula: Quantity = absolute value of Quantity.
7. **Parse mixed dates.** Dates come in two formats, month/day/year (1/6/2023) and day-month-year (06-01-2023). Try the first, then the second; mark anything else as missing.
8. **Check Total.** Formula: Total = Quantity × Unit price × 1.05 (5% tax). If the file's Total is off by more than 0.50, replace it with the formula's value.
9. **Handle what is still missing.** Fill Customer type with "Unknown" and Rating with the median rating. Drop rows with no date, because they cannot be placed in a quarter.
10. **Keep an audit trail.** Write down the row count after each step and how many rows it removed.

Answering the brief

1. **Add time columns.** From Date and Time, make Year, Quarter (e.g. 2023Q1), Day of week and Hour.

2. **Bucket the trading day.** Stores open 10:00–21:00, so use Morning 10–13 (3 hours), Afternoon 13–17 (4 hours) and Evening 17–21 (4 hours). Formula: transactions per hour = transactions in the bucket ÷ hours in the bucket.
3. **Compare against target.** Add up revenue per branch per quarter and join it to a small targets table ($100,000 per branch). Formulas: Variance = Revenue − Target; % of target = Revenue ÷ Target × 100; Hit = Revenue is at least the Target.
4. **Summarise each branch.** Average quarterly revenue, quarters hit out of 8, and best and worst quarter. Formula: YoY growth % = (2024 revenue ÷ 2023 revenue − 1) × 100.
5. **Find the busy times.** Formula: average per day = transactions on that weekday ÷ number of those weekdays in the data. Then do the same for each hour of each weekday and show it as a heatmap.
6. **Chart, export, summarise.** Plot each branch's 8 quarters with a line at $100,000, save the clean data to CSV, and write a five-line summary for management.